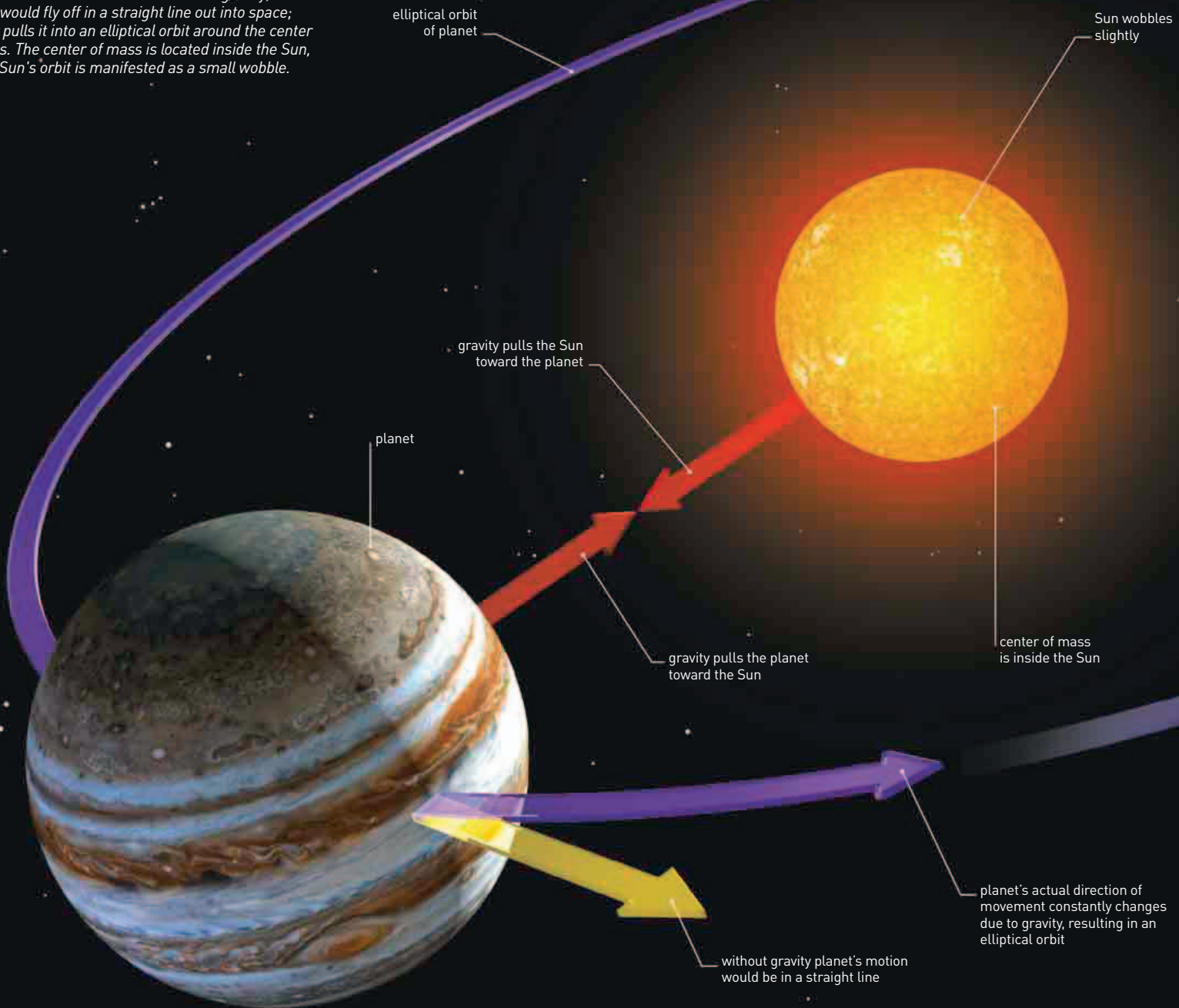


BALANCED FORCES

The Sun exerts a gravitational pull on a planet, and the planet exerts an equal gravitational pull on the Sun. As a result, both the planet and the Sun orbit around a point called the center of mass. Without gravity, the planet would fly off in a straight line out into space; gravity pulls it into an elliptical orbit around the center of mass. The center of mass is located inside the Sun, so the Sun's orbit is manifested as a small wobble.



ORBITAL PERIODS

Kepler's third law gives a mathematical relationship between a planet's average distance from the Sun and its orbital period (the time to complete each orbit). Specifically, it states that the square of the orbital period is proportional to the cube of the semimajor axis (half the diameter of an ellipse at its widest point). This makes it possible to quantify the increase in the orbital period with increasing distance from the Sun. Although Kepler's third law is not as simple as the second law, it enabled Newton to develop his universal law of gravitation.

PLANETARY YEARS

The length of a planet's "year," or orbital period, depends on its average distance from the Sun. The innermost planet, Mercury, has the shortest year at just 88 Earth days. Neptune's is the longest: 60,190 Earth days (164.8 Earth years). The diagram on the right (which is not to scale), shows the planets' orbital periods in Earth years.

